Edmond Sim

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Education

University of Melbourne — Melbourne, VIC
Bachelor of Science (Physics) — *Expected Oct 2027*

- **Standing:** H1 average, WAM 91.55.
- **Awards:** School of Mathematics and Statistics Vacation Scholarship (2026–27); Melbourne National Merit Scholarship; Dean's Honours List; Janet Clarke Hall Academic Honour Roll (2025).
- **Selected results:** Linear Algebra (95, top 15 of 1600+), Real Analysis (93), Vector Calculus: Advanced (91), Quantum and Thermal Physics (97), Physics 1: Advanced (93, ranked 2nd of 100), Chemistry 1 (97, ranked 5th of 1000+).
- **In progress:** Differential Equations, Theoretical Physics 2, Relativity & Electromagnetism, Lab & Computational Physics 2.

Sydney Grammar School — Sydney, NSW
Higher School Certificate — *Graduated 2024*

- **ATAR:** 99.80; HSC All-Rounder. **HSC:** Mathematics Ext. 1 (99, top 10), Physics (97, ranked 2nd), Mathematics Ext. 2 (96).
- **Olympiad:** Australian Physics Olympiad, High Distinction (2023); selected for the 2024 Summer School, top 24 nationally.
- **Co-curricular:** 3rd, Form V Concerto Competition (trombone, Larsson *Concertino*); Principal Trombone, orchestra; recording for Nick Vines's *Knights of Wybornia*; Volleyball (3rds).

Teaching & Work Experience

Head Tutor, Mathematics & Physics — Jul 2025 – Present
Aurum Academy — *Sydney, NSW*

- Teach a cohort of 12 HSC students in Mathematics and Physics, building lessons around spaced retrieval and active recall.
- Write the course materials in LaTeX: Beamer decks for Physics Modules 5 to 8, homework papers with marking guides, topic tests, a Mathematics Extension 1 examination, and a full HSC Physics trial paper, with figures in TikZ.
- Mark student work and write individual feedback against HSC marking criteria.

Exam Marker & Content Developer — Nov 2025 – Present
Fathoming Physics — *Sydney, NSW*

- Mark weekly papers and Year 11 and 12 examinations against rubrics, with written feedback for each student.
- Write mechanics revision booklets in LaTeX and TikZ at two difficulty levels, with worked answers and Moodle XML quizzes.

Private Tutor — Nov 2023 – Present
Self-employed: HSC Mathematics, Physics, English Advanced; Physics Olympiad preparation — *Sydney & Melbourne*

Operations & Technical Support — Jan 2022 – Dec 2024
Dr Rosana Choy Obstetrician/Gynaecologist — *Bankstown, NSW*

- Ran front-office operations (scheduling, insurer claims, payroll, bookkeeping); built the practice website (HTML/CSS/JS).

Computational Work

Variable-Length Pendulum | *Python (NumPy, SciPy, Matplotlib)* — 2026

- Reduced the equation of motion of a pendulum with time-varying length to Bessel's equation; solved and visualised the motion.

Laboratory Data Analysis | *Python, LaTeX* — 2026

- Analysis and write-ups for radiation detection, RC/RLC circuit, and ESR experiments, with curve fitting and error analysis.

Writing & Talks

Feature essays | *OmniSci Magazine (University of Melbourne)* — 2026

- *Through the Looking-Glass* (Issue 11, Oct 2026): Brewster's kaleidoscope as a model of perception, after Locke and Kant.
- *Contingent Realities: the (Ph)ailure of a (Ph)act* (Issue 10, Jun 2026): determinism to relational QM.

Maudlin's Trilemma and the Quantum Measurement Problem | *Physics Student Society seminar* — Sep 2026

- Invited seminar on the measurement problem for a student audience; wrote the Beamer deck and handouts in LaTeX.

University Service

Student Representative, BSc Course Standing Committee — Feb 2025 – Nov 2025
UMSU & University of Melbourne — *Melbourne, VIC*

- Elected to represent 2,100+ students on course structure and academic policy; several of my written submissions were adopted.

Technical Skills

Computing: Python (NumPy, SciPy, Matplotlib, Pandas), MATLAB, HTML/CSS, JavaScript, Flask, SQL (familiar).
Typesetting: LaTeX (Beamer, TikZ, PGFPlots), Moodle XML, Git. | **Languages**: English (native), French (C1), Cantonese.